SLIs, which turns out to be a big advantage, since Prometheus is so widely adopted in a cloud-native context. From an architectural point of view, it comprises the following components:

- As the operator, you use (YAML) documents to define the SLOs and, with it, SLIs.
- A backend watches for `ServiceLevelObjective` resources (either from the local file system or the Kubernetes API) and creates Prometheus recording rules, based on this input.
- The Pyrra API gathers SLIs from a backend, such as Kubernetes.
- The Pyrra UI uses the Pyrra API to display SLOs, SLIs, error budgets, and more.

TIP If you don't want to invest the time needed to set up the Docker-based example here, you can, alternatively, play around with the online environment at demo.pyrra.dev (https://demo.pyrra.dev/).

To stand up the stack using the Docker Compose file shown in listing 10.1, execute the following command from within the `ch10/pyrra/` directory:

```
docker-compose -f docker-compose.yaml up
```

The following listing shows the relevant snippets of the Docker Compose file we use for this exercise, focusing on the configuration of Prometheus and Pyrra.

Listing 10.1 Snippets of the Docker Compose configuration for the Pyrra setup

```
prometheus:
    image: prom/prometheus:v2.40.0
    restart: always
    ports:
      - "9090:9090"
    command:
      - --config.file=/etc/prometheus/prometheus.yml
      - --storage.tsdb.path=/prometheus
      - --storage.tsdb.retention.time=33d
    volumes:
      - ./prometheus/prometheus.yaml:/etc/prometheus/prometheus.yml:ro
      - ./prometheus_pyrra:/etc/prometheus/pyrra

  pyrra-api:
    image: ghcr.io/pyrra-dev/pyrra:v0.5.5
    restart: always
    command:
      - api
      - --prometheus-url=http://prometheus:9090
      - --prometheus-external-url=http://localhost:9090
      - --api-url=http://pyrra-filesystem:9444
    ports:
      - "9099:9099"

  pyrra-filesystem:
    image: ghcr.io/pyrra-dev/pyrra:v0.5.0
```

Tells Prometheus where the rule files are stored locally

Configures Pyrra with the Prometheus info

Sets up Pyrra's local file system provider (to store generated rules)